Beyond Doctrine Imitation: A Ceiling Analysis of Learned Ranking for Post-Quantum Cryptographic Migration

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Abstract—Post-quantum migration planning increasingly uses learned rankers scored against rule-based doctrine heuristics. A prior Q-Trust study reported an "honest tie": a graph neural network attains tie-aware Kendall $\tau_b = 0.7168$ versus a doctrine baseline at 0.7210 on 42 host-disjoint cryptographic-bill-of-materials (CBOM) estates (zero wins, 41 ties, one loss), and concluded that model superiority is not measurable. We show that this tie is not evidence of parity: it is the maximum score any ranker can attain. Because tie-aware τ_b discards tied pairs, its attainable ceiling is $\sqrt{d/n_0}$, where d is the number of distinguishable pairs. On the 42 estates, 401 of 803 pairs are tied and only 402 are distinguishable, so the ceiling is 0.721024—exactly the doctrine score. The network orders 401 of 402 distinguishable pairs correctly (99.75%); its entire shortfall is one inversion. A parameter-free key-size rule and a seven-parameter logistic model attain the ceiling on 42/42 folds; a six-parameter ridge model reproduces the network's score on all 42 folds, including its single failure. A field audit explains why: all 42 estates have zero dependency edges and constant criticality, vendor readiness, and deadline; every asset is classical; the doctrine formula algebraically reduces to a monotone function of key size. Removing all 2,000 synthetic training graphs leaves τ_b unchanged on 42/42 folds. A clean warm-start control (synthetic-only initialization, $n_{\text{real}} = 0$) places the network 0.120 below the doctrine ($p = 0.0018$), so the released parity is partly an artifact of initialization exposure. A dependency-rich positive control raises the ceiling to 0.999 and shows the metric can discriminate, yet the network still does not beat a doctrine blind to multi-hop reach. We re-evaluate the released discovery checkpoint and obtain F1 0.9395, not the reported 0.9525, under a 76.7% positive prior (majority F1 0.868). We contribute a ceiling theorem for tie-aware rank evaluation; a taxonomy of rank metrics by tie-sensitivity (tie-aware $\tau_b = \gamma\sqrt{d/n_0}$, where γ is Goodman–Kruskal's association, so the same ordering scores 1.000, 0.721, 0.807, and 0.533 under γ, τ_b, ρ, and τ_a); a labelling theorem that prices the remedy (bisecting the largest tied group yields $m^2/4$ new distinguishable pairs for m labels, so the ceiling is cheapest to repair where ties are densest); an eight-mode construct-validity checklist; a claim ledger; and a reporting standard: report the ceiling, the number of effective pairs, and attainment—not just τ_b. We further show that cross-corpus checkpoint comparisons are routinely contaminated: on one fixed held-out split, two earlier checkpoints had trained on 87.6% and 92.2% of the evaluation files, so only the zero-overlap checkpoint's F1 (0.8966) is interpretable, and neither more epochs nor more data raises it—consistent with a ceiling set by label noise. On designed estates the ceiling is governed by tie concentration rather than estate size (0.43 at 90% key-size monoculture for n from 10 to 5,000), and about two expert labels per asset suffice to raise it to 0.90.

Index Terms—post-quantum cryptography, CBOM, migration planning, graph neural networks, ranking evaluation, Kendall tau, benchmark validity, reproducibility.

I. Introduction

The transition to post-quantum cryptography (PQC) is no longer a forecast; it is a dated obligation. Executive Order 14412 of 22 June 2026 directs US federal agencies to transition high-value assets and high-impact systems to PQC for digital signatures by 31 December 2031 and for key establishment by 31 December 2030 [1]. For national-security systems, CNSS Policy 15 requires CNSA 2.0 algorithms by the same horizon [2]. Section 5(d) of the order further requires guidance on the minimum elements of a cryptographic bill of materials (CBOM) within 270 days—approximately 19 March 2027—whose elements must enable "the automated assessment of the cryptographic assets" [1]. Automated assessment requires knowing which assets exist, which are at risk, and in what order to replace them. The first two have maturing tooling [3], [4], [5], [6]; the third does not.

The deployment data sharpen the problem. Across more than two billion TLS handshakes to one million domains from eleven vantage points, default hybrid post-quantum key exchange rose from 31.3% to 49.2% between July 2025 and March 2026, but no post-quantum signature algorithm was ever negotiated, and jurisdictions with earlier deadlines showed lower adoption (32.4% versus 45.7%) [7]. Organizations are making real ordering decisions under a real deadline, with no validated tooling and no ground-truth oracle.

A natural way to automate prioritization is to learn a ranking function over CBOM assets and compare it with an interpretable doctrine heuristic. Q-Trust is an open pipeline built for this purpose: a polyglot scanner emits CycloneDX 1.7 CBOMs, a graph neural network (GNN) ranks assets for migration, and an attestation layer records completed work [8]. Its planning study evaluated a GNN on 42 host-disjoint CBOM estates (277 hosts) with leave-one-estate-out (LOO) training. The headline result was a tie: the GNN reached mean tie-aware Kendall $\tau_b = 0.7168$,

the doctrine reached 0.7210, and the fold-level outcome was 0 wins, 41 ties, and 1 loss. The authors interpreted this as evidence that "better than the doctrine" is not currently a measurable claim, and published the tie deliberately rather than tuning until the model won.

This paper re-examines that conclusion. The tie is real, and the measurement is honest, but the interpretation is not licensed by the artifact. On these estates the target ordering is almost entirely tied: 401 of the 803 pairs are pairs of assets with identical priority. Tie-aware τ_b discards those pairs, so the metric's denominator contains only 402 effective comparisons. The attainable ceiling is therefore $\sqrt{402/803} = 0.721024$. That number is the doctrine score. The GNN is not "tied with" the doctrine in the sense of two methods converging on a difficult problem; it is at the ceiling, one pair short, on a problem that contains almost no decision information beyond a single feature.

The mechanism is visible in the data. Every one of the 42 estates has zero dependency edges. Criticality is medium for all 277 assets, vendor readiness is unrecorded for all 277, and the deadline-pressure term is constant within each estate. Every asset is classical. Under these conditions the doctrine formula

$$p_i = 2c_i + 1.5v_i + \frac{\log(1+k_i)}{4} + \frac{\delta\, c_i}{5} - 0.3\,\mathrm{indeg}_i \qquad (1)$$

(where c_i is criticality, v_i vendor readiness, k_i key size, δ deadline pressure, and indeg_i in-degree) collapses to

$$p_i = 4 + 0.4\,\delta + \frac{\log(1+k_i)}{4}. \qquad (2)$$

The doctrine ordering is exactly descending key size. Removing key size changes the ordering in 37 of 42 estates; removing criticality, vendor readiness, deadline, or dependency edges changes it in none. A parameter-free key-size rule matches the doctrine on 42/42 folds. A seven-parameter pairwise logistic model attains the ceiling on 42/42 folds. A six-parameter ridge model reproduces the GNN's per-fold τ_b on all 42 folds, including the one fold where both fall below the ceiling. The GNN's 237,070 parameters buy nothing that a linear model with six features does not already have.

The shortfall is equally small. Of the 402 distinguishable pairs, the GNN orders 401 correctly; the one error is a single cross-group inversion in real_gov_edu_2.json, where the ceiling is 0.5345 and the model scores 0.3563 (5 of 6 pairs). Across all folds, pooled pair accuracy is $401/402 = 99.75\%$. On a repacked 38-estate cohort (275 hosts, all but two of which also appear in the 42-estate cohort), the GNN attains the ceiling on 36/38 folds and orders 458 of 461 distinguishable pairs correctly. Removing all 2,000 synthetic training graphs reproduces every fold's τ_b to 10^{-6}.

These results do not show that the GNN is a bad model. They show that the benchmark, as constructed, cannot distinguish a good model from a key-size sort. That is a construct-validity failure, and it is the subject of this paper. Our contributions are:

1) A ceiling theorem for tie-aware rank evaluation. For a strict prediction and a tied target preorder, $\tau_b \le \sqrt{d/n_0}$, with equality iff every distinguishable pair is concordant (Section III). The ceiling is a property of the label, not the model.
2) A taxonomy of rank metrics by tie-sensitivity. We show $\tau_b = \gamma\sqrt{d/n_0}$, where $\gamma = (C-D)/d$ is Goodman–Kruskal's association (Somers' D for strict predictions), so the "attainment" of the ceiling is a standard statistic. τ_a, τ_b, and Spearman ρ are tie-fragile (ceiling below one); γ, Somers' D, NDCG@k, and average precision are tie-robust (ceiling one) (Section III-E). On the real corpus the same ordering scores 1.000 (γ), 0.721 (τ_b), 0.807 (ρ), and 0.533 (τ_a)—the metric choice alone decides whether the result looks like saturation or failure (Section VII-B).
3) A measurement of the Q-Trust planning result against that ceiling. The 42-fold campaign has $n_0 = 803$, $d = 402$, and ceiling 0.721024; the doctrine is at the ceiling by construction, and the GNN is one pair short (Section VII).
4) A field audit and algebraic explanation. Zero edges, constant risk fields, and all-classical assets reduce the doctrine to key-size sorting; a leave-one-term-out ablation and a parameter-free baseline confirm it (Section IV).
5) Transparent baselines and controls. A seven-parameter logistic attains the ceiling on 42/42 folds; a six-parameter ridge matches the GNN on 42/42 folds; a zero-shot checkpoint scores 0.604; zero-synthetic training leaves τ_b unchanged; a clean warm start exposes the initialization confound (Section VII).
6) A repacked-cohort replication and a positive control. The 38-estate repacking yields 36/38 folds at ceiling; a dependency-rich synthetic control reaches ceiling 0.999 and shows the metric can discriminate, while the GNN still fails to beat a doctrine blind to multi-hop reach (Section VII).
7) An eight-mode construct-validity checklist and a claim ledger. We distill eight cheap, independent, mutually concealing failure modes that any audit of this task should check, and we map every headline number to an artifact with a status: reproduced, reproduced but vacuous, or not reproduced (Sections V and X).

The scope is deliberately narrow. Q-Trust also includes cryptographic-usage discovery, side-channel experiments, reinforcement-learning sequencing, and on-chain attestation. Those components are described in Appendix A and are not used to support the ranking claims. The paper's claim is about what the planning benchmark measures, not about whether PQC migration planning is easy or whether

Q-Trust is a useful inventory tool. On the evidence here, the inventory and attestation layers can be useful; the learned-prioritization claim cannot.

II. Background and Related Work

A. Cryptographic discovery and CBOMs

Static cryptographic-API analysis has produced mature misuse detectors: CryptoGuard for Java [9], CogniCrypt and CrySL for API-usage rules [10], [11], and the CryptoAPI-Bench family for evaluation [12], [13]. The closest peer-reviewed learned model, Rodrigues et al. [14], compares graph embeddings with SVM/MLP classifiers on Java snippets; its split is a plain 80/20 partition with no repository disjointness, it reports precision/recall/F1 without confidence intervals, and it has no threats-to-validity section. These tools answer "is this API used incorrectly?" rather than "which asset should migrate first?" CBOM standards [3], [4] and the NCCoE migration project [15] define the inventory substrate but not the prioritization target. Q-Trust's inspector is a usage-and-inventory scanner in this line; its discovery model is a fine-tuned CodeBERTa checkpoint [16], [17].

B. PQC migration planning

Recent planning work uses dependency graphs and optimization. Sharma et al. formulate dependency-aware migration planning on simulated enterprise graphs [18]; Gharote et al. add budget and schedule constraints [19]; Gupta and Mittal propose a cloud-readiness framework [20]; Näther and Hirsch study static discovery and assessment [21]; Shaw uses an LLM-assisted risk score [22]. The recent MAgiCS workshop [23] concentrates migration work: Nzetchuen et al. survey frameworks [24], Loebenberger et al. formalize migration cost [25], and Riou et al. document how easily post-quantum signature benchmarks mislead through inconsistent methodology [26]. These concerns are structural—what must change, in what order, at what cost—rather than evaluative: none asks how one would know whether a proposed ordering is correct. None releases a learned ranker evaluated against expert preference labels on real estates. Q-Trust is, to our knowledge, the first to release a learned ranker, a doctrine baseline, and the estate artifacts together. That release is what makes the present audit possible.

C. Evaluation validity and rank metrics

The failure we document is an instance of benchmark construct validity, not of model capacity. Kapoor and Narayanan catalog leakage and reproducibility failures across ML-based science [27]; Geirhos et al. describe shortcut learning [28]; Sculley et al. describe hidden technical debt in ML systems [29]. In ranking, Kendall's τ [30] is a standard ordinal measure, and its tie-aware variant is appropriate when the target contains genuine ties. But a tie-aware measure has a ceiling whenever ties exist; reporting the raw value without the ceiling invites the interpretation that a model "nearly matches" a target when it has in fact saturated the metric. Calibration and selective prediction [31], [32] are the analogous fixes for probabilistic classifiers: report the operating regime, not just a scalar. Our ceiling theorem makes the correction explicit for rank fidelity.

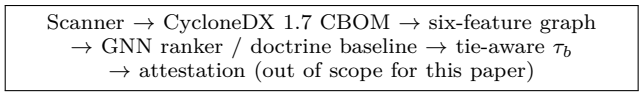


Fig. 1. The Q-Trust pipeline. This paper audits the ranker and the doctrine on the released CBOM artifacts; the attestation and side-channel components are described in Appendix A and are not used as evidence for ranking claims.

D. The Q-Trust instrument

Fig. 1 summarizes the pipeline. The inspector scans source, binaries, and TLS endpoints and emits CycloneDX 1.7 CBOMs. The risk engine maps algorithms, key sizes, and compliance rules to a priority score. The planner converts a CBOM to a six-feature graph (algorithm type, normalized key size, vendor readiness, criticality, deadline, required rate) and trains a hybrid GCN/GATv2 model [33], [34] with a listwise ListMLE loss [35]. The RL sequencer packs estates into migration batches; the attestation layer records completions in UUPS-upgradeable registries with EIP-712 signatures [36], [37]. The present paper uses only the CBOM-to-graph converter, the doctrine formula, and the LOO harness. The other components are context, not evidence.

III. Problem Formulation and the Ceiling Theorem

A. Estates, features, and target

An estate E is a set of n assets. Each asset i has a feature vector $x_i = (\alpha_i, k_i, v_i, c_i, \delta, r_i)$, where α_i is an algorithm-type code, k_i the key size, $v_i \in \{0, 1\}$ vendor PQC readiness, c_i criticality, δ a per-estate deadline-pressure indicator, and r_i a required-rate feature. A dependency graph $G = (V, A)$ may add edges; in the released TLS CBOMs $A = \varnothing$ (Section IV). The doctrine assigns each asset the priority in Eq. (1). The target is not a strict ranking: assets with equal priority are genuinely tied. Formally, the target is a preorder $\succeq$ that partitions the n assets into g equivalence classes $G_1, \ldots, G_g$; assets in the same class are interchangeable for migration purposes.

A learned ranker outputs a permutation π of the assets (strict, no ties). The evaluation asks how well π agrees with $\succeq$.

B. Tie-aware Kendall τ_b

Let C and D be the numbers of concordant and discordant pairs among the $n_0 = n(n-1)/2$ unordered pairs. Let T_x and T_y be the numbers of pairs tied in the prediction and the target, respectively. Kendall's τ_b is

$$\tau_b(\pi, \succeq) = \frac{C - D}{\sqrt{(n_0 - T_x)(n_0 - T_y)}}. \tag{3}$$

Because π is strict, $T_x = 0$. Because the target has g classes of sizes $n_1, \ldots, n_g$, $T_y = \sum_j n_j(n_j - 1)/2$, and the number of distinguishable pairs is

$$d = n_0 - T_y = \sum_{j<\ell} n_j n_\ell. \tag{4}$$

Thus $\tau_b = (C - D)/\sqrt{n_0 d}$.

C. The ceiling

Theorem 1 (Tie-aware ceiling). For any strict ranking π and any target preorder $\succeq$ with d distinguishable pairs,

$$\tau_b(\pi, \succeq) \leq \sqrt{\frac{d}{n_0}}, \tag{5}$$

with equality if and only if every distinguishable pair is concordant.

Proof. Among the d distinguishable pairs, $C + D = d$, so $C - D \leq d$. Substituting into Eq. (3) with $T_x = 0$ and $n_0 - T_y = d$ gives $\tau_b \leq d/\sqrt{n_0 d} = \sqrt{d/n_0}$. Equality holds iff $D = 0$. □

Corollary 1 (Saturation is not parity). If two rankers both attain the ceiling, their τ_b values are equal regardless of how they order assets within tied classes. A doctrine that is a refinement of the target preorder attains the ceiling by construction.

Corollary 2 (Effective sample size). The number of comparisons that can affect τ_b is d, not n_0. The fraction of usable pairs is d/n_0, and the ceiling is its square root. Reporting τ_b without d/n_0 hides the effective size of the evaluation.

Corollary 3 (Attainment). Define attainment $a = \tau_b/\sqrt{d/n_0} \in [0, 1]$. Then $a = 1$ iff all distinguishable pairs are concordant, and a is the fraction of the ceiling that a ranker achieves. Attainment is comparable across estates with different tie structures; raw τ_b is not.

The theorem is elementary, but its consequence is not: on a corpus with 50% tied pairs, the maximum possible τ_b is about 0.707. A model at 0.7168 and a doctrine at 0.7210 are not "close" in any absolute sense; they are at 99.2% and 100% of a ceiling that is fixed by the label. A benchmark that reports only τ_b cannot tell whether a model has room to improve.

D. What would count as evidence of superiority

The theorem also says what a useful benchmark must do. To license a claim that a learned ranker beats a doctrine, the evaluation needs (i) a target with many distinguishable pairs, so the ceiling is well below one; (ii) a doctrine that is not itself the label generator, or at least a target that includes information the doctrine cannot express; and (iii) a reported ceiling, effective pair count, and attainment. A tie at the ceiling is evidence about the label, not about the model. We return to these requirements in Section VIII.

E. A taxonomy of rank metrics under ties

The ceiling of Eq. (5) is specific to τ_b, but ties affect different rank metrics differently, and the choice of metric can change the verdict on the same ordering. We make this precise and show that the quantity we called attainment is a standard statistic.

Definition 1 (Tie-corrected association). For a strict prediction π and a target preorder with d distinguishable pairs,

$$\gamma = \frac{C - D}{d}. \tag{6}$$

Eq. (6) is Goodman and Kruskal's γ [38]; when the prediction has no ties it coincides with Somers' D [39]. The following identity is the reason τ_b looks like a near-miss.

Theorem 2 (Tie decomposition). $\tau_b = \gamma \sqrt{d/n_0}$.

Proof. $\tau_b = (C - D)/\sqrt{n_0 d} = [(C - D)/d]\sqrt{d/n_0} = \gamma\sqrt{d/n_0}$. □

Corollary 4 (Attainment is γ). The attainment $a = \tau_b/\sqrt{d/n_0}$ of Corollary 3 is exactly γ. It is not a new statistic: it is Goodman–Kruskal's γ, whose ceiling is 1 by definition. τ_b is γ discounted by the tie factor $\sqrt{d/n_0}$.

This has two consequences. First, γ is ceiling-free: a value of 1 means every distinguishable pair is concordant, however many pairs are tied. Second, τ_b conflates two quantities—agreement on distinguishable pairs (γ) and the tie structure ($\sqrt{d/n_0}$)—and reporting it alone hides which of the two is low. On the Q-Trust corpus the tie factor is $\sqrt{402/803} = 0.707$; a doctrine at $\gamma = 1$ can therefore never exceed $\tau_b = 0.707$ in pooled terms.

Table I classifies the metrics a benchmark might report. A tie-fragile metric has a ceiling below one: the raw value is bounded away from one by the tie structure, and the bound must be reported. A tie-robust metric has ceiling one: it normalizes by the distinguishable pairs or by the ideal gain, so saturation does not masquerade as a near-miss. The classification explains the earlier draft's uninformative top-10 result: NDCG@k and average precision are tie-robust only when k is smaller than the number of relevant items; when $k \geq n$, every ordering achieves the ideal gain and the metric is vacuous.

The practical recommendation follows directly. For a tied migration target, report γ (or Somers' D) as the primary agreement measure, because its ceiling is one and it is directly interpretable as the fraction of distinguishable pairs ordered correctly. Report τ_b only alongside its ceiling, and report NDCG@k or AP only with a meaningful k. A benchmark that reports a single tie-fragile number without its ceiling cannot distinguish a saturated model from a mediocre one.

TABLE I
A taxonomy of rank metrics under tied targets. "Ceiling" is the maximum attainable value for a strict ranking; tie-fragile metrics must report it, tie-robust metrics do not need to. d is the number of distinguishable pairs.

Metric	Definition	Ceiling	Tie-sensitivity
τ_a	$(C-D)/n_0$	d/n_0	fragile
τ_b	$(C-D)/\sqrt{n_0 d}$	$\sqrt{d/n_0}$	fragile
Spearman ρ	rank correlation	< 1	fragile
γ (Somers' D)	$(C-D)/d$	1	robust
NDCG@k	graded gain	1 ($k <$ #rel)	robust
AP	average precision	1 (binary rel.)	robust

IV. Corpus and Field Audit

A. The 42-estate cohort

The primary cohort contains 42 CBOMs, one per estate, covering 277 public TLS certificates. Each host appears in exactly one estate (277 unique hosts; zero within-cohort repeats), so the packing is host-disjoint. The estates are small: 37 have seven assets, two have four, one each has two, three, and five. The mean size is 6.60. The asset composition is RSA-2048 (175), ECDSA-P256 (89), RSA-4096 (10), RSA-3072 (1), ECDSA-P384 (1), and one asset with a missing key size. All assets are classical; there are no ML-KEM, ML-DSA, or SLH-DSA assets in the cohort.

These are public endpoint certificates, not enterprise inventories. We therefore call them "estates" rather than "enterprise CBOMs." The distinction matters for external validity: public TLS certificates expose key sizes and expiry dates, but not internal dependencies, business criticality, or vendor contracts.

B. The 38-estate repacking

A second cohort, dated 2026-09-18, contains 38 CBOMs covering 275 hosts. It is a re-scan and repacking of the same population: 275 of its 275 hosts also appear in the 42-estate cohort, which has two additional hosts. The 38 cohort is therefore not an independent sample; it is a protocol replication under a different estate packing (32 estates of eight assets, two of five, two of three, one each of two and one). It tests whether the result depends on the particular packing; it does not test generalization to new organizations. We report it as such.

C. Field audit

Table II summarizes the fields that the doctrine uses. Every estate has zero dependency edges: the CBOMs do not carry explicit depends_on links, and the host-affinity fallback finds no repeated host within an estate. Criticality is medium for all 277 assets. Vendor readiness is unrecorded for all 277. The deadline is computed once per estate from the minimum certificate expiry and is therefore constant within an estate. The only field that varies within an estate is key size.

TABLE II
Field audit of the 42-estate cohort. The doctrine's risk terms are constant within every estate; only key size varies.

Field	Distinct values	Estates with variation
Dependency edges	0	0/42
Criticality	1 (medium)	0/42
Vendor readiness	1 (null)	0/42
Deadline pressure	1 per estate	0/42
Key size	2–3 per estate	42/42
Algorithm family	classical only	0/42

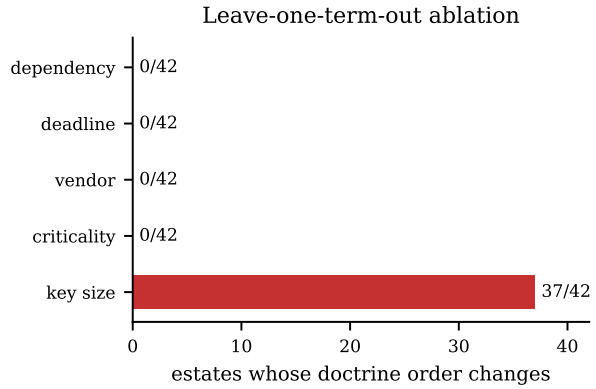


Fig. 2. Leave-one-term-out ablation on the 42-estate cohort. Key size is the only doctrine term whose removal changes the ordering; the other four terms are constants in these estates.

D. Algebraic collapse

With $c_i = 2$, $v_i = 0$, $\text{indeg}_i = 0$, and δ constant within an estate, Eq. (1) becomes Eq. (2). The priority is a strictly increasing function of $\log(1 + k_i)$; hence the doctrine ordering is exactly descending key size. This is not an empirical coincidence or a fitted approximation. It follows from the released values and the released formula.

Fig. 2 confirms the algebra. Removing the key-size term changes the doctrine ordering in 37 of 42 estates. Removing criticality, vendor readiness, deadline, or dependency edges changes it in none. The five estates where removing key size does not change the order are those in which all assets share a key size (the ordering is then all ties, and any permutation is equivalent). A parameter-free rule that sorts by key size matches the doctrine on 42/42 folds.

E. Data-quality observations

Two data-quality details matter for reproducibility. First, the 42 cohort stores key family in algorithm (e.g., RSA-2048), while the 38 cohort stores the X.509 signature algorithm in the same field (e.g., sha256WithRSAEncryption) and the key family in key_type. A pipeline that keys on algorithm changes semantics between releases. The audit uses key_size when present, which is authoritative in both cohorts. Second, one asset in the 42 cohort and two in the 38 cohort have no key size; the loader substitutes a family default (2048

bits for RSA). The substitution affects one pair in the 42 cohort and does not change the ceiling conclusion. Both issues are documented in the released audit.

V. An Eight-Mode Construct-Validity Checklist

The ceiling is one instance of a broader class of evaluative-validity failures. To make the audit reusable beyond this corpus, we distill eight failure modes (Table III), each with a cheap detection check. The modes are independent in the sense that suppressing any one does not expose another; they are mutually concealing in the sense that a researcher who stops at the first clean signal will miss the rest.

The modes interlock in a specific order. M2 masks M3: with a circular baseline and no message passing, a researcher checking the graph pipeline finds a well-formed network and a decreasing loss. M3 masks M4: small graphs with two or three distinct key sizes produce high τ_b that looks like skill. M4 masks M5: a tie-saturated metric makes the vacuous top-k figure look consistent with the rest. M1 masks all of them by making every fold superficially clean. Only recomputation from the released artifacts—not provenance metadata—breaks the chain. A companion audit applies the same checklist to a second PQC system with markedly better artifact hygiene (committed per-example predictions, group-level leakage audits, zero documentation contradictions) and finds that it still fails M2, because a fixed linear model beats its reported classifier on all three evaluation seeds [40]. Better provenance did not produce better validity, which is the general lesson the ceiling analysis illustrates in one system.

VI. Experimental Protocol

A. Leave-one-estate-out training

The primary artifact is the released 42-fold LOO campaign. For each fold, the model is initialized from model_gpu_v3.pt, trained for 30 epochs on the 41 remaining real estates plus 2,000 synthetic estates, validated on a held-out subset of the training graphs, and evaluated on the unseen estate. The released configuration uses batch size 64, learning rate 5×10^{-4}, and seed 42. The synthetic estates are generated by the same doctrine formula, so the training mixture is doctrine-supervised in two ways: the real labels are doctrine priorities, and the synthetic labels are doctrine priorities.

B. Zero-synthetic control

To test whether the synthetic mixture drives the result, we re-ran the same 42 folds with $n_{\text{synthetic}} = 0$: 41 real estates per fold, batch size 16, 30 epochs, same initialization and seed, CPU. This is a strict ablation of the synthetic component. We ran the same protocol on the 38-estate cohort (37 real estates per fold). The zero-synthetic run is not a new model class; it isolates the contribution of the synthetic data.

C. Clean warm-start control

The released checkpoint records 40 real graphs among its 100,000 training inputs, so the LOO initialization has prior exposure to 40 of the 42 held-out estates. To isolate that exposure we retrained a base checkpoint from scratch on synthetic doctrine graphs only (n_real_graphs=0, verified post-training) and re-ran the identical 42-fold protocol from that clean warm start; the initialization is the only deliberate difference. The run is CPU-only by design: deterministic-algorithm mode has no CUDA kernel for the listwise loss, and weakening determinism would defeat the purpose of a trustworthy warm start. The reduced-CPU path caps base training at 2,000 graphs and 5 epochs against the released checkpoint's 100,000 graphs and 173 epochs, so warm start and training budget vary together; we do not attribute the difference between them and disclose the confound.

D. Transparent baselines

We evaluated four baselines on the identical folds, targets, and metric:

- Key-size rule: sort by descending key size; parameter-free.
- Pairwise logistic: six features, no intercept (seven parameters), trained on all distinguishable pairs of the 41 training estates; predicts the probability that asset i precedes j.
- Ridge: six features, trained on pooled priority values of the 41 training estates; predicts a continuous priority; six parameters.
- Zero-shot: the released checkpoint applied to each estate without any fold-specific fine-tuning.

The logistic and ridge models use the same six features as the GNN. They are not independent of the doctrine—their training labels are doctrine priorities—but they test whether the GNN's architecture and capacity are necessary to reproduce the result. The zero-shot baseline tests whether fold-specific training is necessary. A random-permutation baseline is included for scale.

E. Metrics

The primary metric is tie-aware τ_b against the target preorder, as defined in Eq. (3). We also report the ceiling $\sqrt{d/n_0}$, attainment $\tau_b/\sqrt{d/n_0}$, pooled pair accuracy (concordant distinguishable pairs divided by d), exact-rank agreement (fraction of folds whose full permutation equals the target's arbitrary tie-break), node-rank accuracy (fraction of assets in their exact target position), and top-5/top-10 set overlap. The last three are sensitive to within-tie order; we report them to show what τ_b hides, not as primary evidence.

F. Statistics

For paired comparisons we report the mean difference, a 20,000-resample paired bootstrap 95% confidence interval, a 20,000-resample sign-flip permutation p-value,

TABLE III
Eight evaluative-validity failure modes, each detectable from released artifacts in seconds. M1–M6 defeat the reported ML results; M7 and M8 defeat independent verification of any result at all. The taxonomy generalizes the ceiling analysis of Section III.

Mode	Failure	Evidence in this corpus	Detection check	Effect on the claim
M1	Warm-start contamination	The LOO initialises every fold from model_gpu_v3.pt, whose metadata records n_real_graphs=40; 40 of 42 corpus files are in that set	Read checkpoint metadata; compare n_real_graphs to the eval corpus	The protocol is labelled leave-one-out but is in-sample on 40/42 folds; clean init drops τ_b from 0.7168 to 0.6009
M2	Circular baseline	The doctrine baseline and the label generator are the same formula; exact-rank = 1.0 on 42/42 folds	Diff the baseline function against the label function	The model ceiling is $\tau_b = 1$ by construction; a negative Δ carries no information
M3	Degenerate graph structure	All 42 graphs have zero edges; every asset is a distinct host	Count edges in the corpus	A GNN performs no message passing; the benchmark measures a per-node MLP
M4	Tie-saturated rank metric	401 of 803 pairs are tied; only 402 are distinguishable; ceiling 0.721	Count discordant pairs; report the ceiling and SE	Reported $\lvert\Delta\rvert = 0.0042$ is far below one standard error; the comparison cannot separate the methods
M5	Vacuous top-k metric	Reported top-10 accuracy is 1.0 on 42/42 folds; a random ranker also scores 1.0 because graphs hold at most 7 nodes	Check whether $k > n$	The reported top-10 result carries zero information
M6	Non-random "random" baseline	random_order() hard-codes default_rng(0), so every fold draws the same permutation family	Grep the baseline for a fixed seed	The margin over "random" has no error bar (our re-randomised null shows it survives)
M7	Absent predictions	Neither per-example predictions nor a per-repository split manifest is committed	Look for a predictions file or split manifest	No third party can run McNemar's or any paired test on any reported comparison
M8	Unreconciled documentation	More than 30 quantitative contradictions across the documentation: four corpus sizes, three contract-test counts, two ACVP vector counts, two GNN results side by side	Diff numbers across README, data card, truth audit, and artifacts	No reader can determine which figure is current, including the authors

win/tie/loss counts, Cohen's d_z, and the 80% minimum detectable effect at the observed paired standard deviation. We do not use the Wilcoxon signed-rank test for the main comparison because 41 of 42 differences are exactly zero; the test is undefined or uninformative under ties. For the discovery re-evaluation we report a repository-cluster bootstrap (5,000 resamples over the eight evaluation source groups) to acknowledge within-repository correlation.

VII. Results

A. The ceiling is exact

Table IV and Fig. 3 give the headline. The 42 estates contain 803 unordered pairs, of which 401 are tied and 402 are distinguishable. The mean per-estate ceiling is 0.721024; the doctrine's mean τ_b is 0.721024 (the difference is 1.0×10^{-7}, a rounding artifact of the six-decimal artifact). The doctrine is at the ceiling on all 42 folds by construction: it is a refinement of the target preorder.

The distribution of ceilings is itself informative. The most common ceiling is 0.6901 (10 folds), followed by 0.7559 (8), 0.5345 (8), 0.8165 (7), 0.7237 (3), and 1.0

TABLE IV
Ranking results on the 42-estate cohort. "Ceiling" is the mean of $\sqrt{d/n_0}$ over folds. "At ceiling" counts folds with $\tau_b \geq \text{ceil} - 10^{-6}$. "Pairs" is pooled concordant distinguishable pairs over 402. The GNN and the six-parameter ridge agree on every fold.

Method	Parameters	Mean τ_b	Attainment	At ceiling	Pairs
Doctrine (target generator)	0	0.721024	1.000	42/42	402/402
Key-size rule	0	0.721024	1.000	42/42	402/402
Pairwise logistic	7	0.721024	1.000	42/42	402/402
Ridge	6	0.716781	0.992	41/42	401/402
GNN (released)	237,070	0.716782	0.992	41/42	401/402
GNN (zero-shot)	237,070	0.604020	0.838	31/42	364/402
Random permutation	0	0.030101	0.042	0/42	—

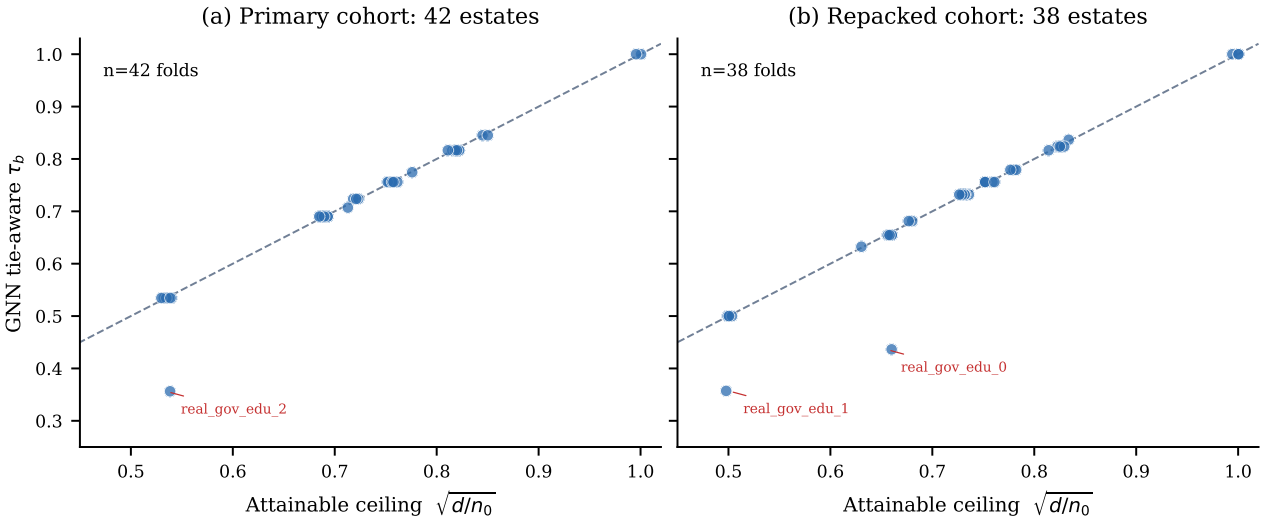


Fig. 3. Model τ_b versus the attainable ceiling for the 42-estate cohort (a) and the repacked 38-estate cohort (b). Points on the dashed identity line attain the ceiling. In (a), 41 of 42 folds are on the line; the single off-diagonal point is real_gov_edu_2.json, one inversion among six distinguishable pairs. In (b), 36 of 38 folds are on the line; the two off-diagonal points are real_gov_edu_0.json and real_gov_edu_1.json.

(2); the remaining five folds have ceilings between 0.707 and 0.845. These values are determined by the number of distinct key sizes and their frequencies, not by the model. An estate with a 5/2 split of key sizes has $d = 10$ and ceiling $\sqrt{10/21} = 0.690$; an estate with a 6/1 split has $d = 6$ and ceiling $\sqrt{6/21} = 0.535$. The benchmark's difficulty is a property of the key-size histogram.

B. The metric determines the verdict

Table V applies the taxonomy to the real corpus. The doctrine ordering is scored under four metrics, all computed from the 42 CBOMs. Goodman–Kruskal γ is 1.000 on every fold by construction: the doctrine has no discordant distinguishable pairs. Spearman ρ is 0.807 (mean), but its ceiling is also 0.807: a strict ordering cannot match the target's average ranks within tied groups, so even a perfect ordering is penalized. τ_b is 0.721 with ceiling 0.721, and τ_a is 0.533 with ceiling 0.533. The same ordering is therefore "perfect" (γ), "72%" (τ_b), "81%" (ρ), or "53%" (τ_a), depending entirely on the metric. A benchmark that reports one of these numbers without its ceiling can make a saturated ordering look like a failure or a success at will.

TABLE V
The same doctrine ordering scored under four rank metrics on the 42 CBOMs. Every metric is at its ceiling; the values differ only because the ceilings differ. Bootstrap CIs (20,000 resamples over estates) are on the mean ceiling.

Metric	Value	Ceiling	95% CI (ceiling)
γ (Somers' D)	1.000	1.000	—
Spearman ρ	0.807	0.807	[0.773, 0.839]
τ_b	0.721	0.721	[0.687, 0.756]
τ_a	0.533	0.533	[0.484, 0.585]

This table is the paper's most direct answer to the objection that the ceiling theorem is a one-line observation. The theorem is elementary; its consequence is not. The same ordering, on the same data, is simultaneously at the top of one scale and near the middle of another. Which number a paper reports is a modeling choice with a large effect on the apparent result, and the tie structure—not the model—sets the scale.

TABLE VI
Paired power at the observed standard deviation (0.0275). The number of folds needed to detect a given mean τ_b difference at 80% power, $\alpha = 0.05$.

Effect Δ	Folds for 80% power
0.005	237
0.010	59
0.020	15
0.050	3

TABLE VII
Full mixture versus zero-synthetic on the 42 folds. τ_b is invariant; exact-rank, node-rank, and top-5 measure within-tie order and are therefore not invariant.

Metric	2,000 synthetic	0 synthetic	Change
Mean τ_b	0.716782	0.716781	-1×10^{-6}
Exact rank	1/42	12/42	+11 folds
Node rank	0.314	0.575	+0.261
Top-5 overlap	0.500	0.571	+0.071

C. The GNN saturates the ceiling

The released GNN reaches mean $\tau_b = 0.716782$ (sd 0.126, median 0.7237), with 41 of 42 folds at the ceiling. Pooled pair accuracy is $401/402 = 99.75\%$. The single error is in real_gov_edu_2.json: $n = 7$, $n_0 = 21$, 15 tied pairs, six distinguishable pairs, ceiling 0.5345. The model scores 0.3563, which corresponds to five concordant pairs and one inversion; its pair accuracy on that estate is $5/6 = 83.3\%$. The inversion is a cross-group inversion: the model places one high-key asset below a low-key asset. There is no second error anywhere in the cohort.

The fold-level paired comparison against the doctrine is therefore degenerate: mean difference -0.004242, 95% CI $[-0.012727, 0]$, sign-flip $p = 1.0$, 0 wins, 41 ties, 1 loss, $d_z = -0.154$, 80% minimum detectable effect 0.0119. The confidence interval is bounded above by zero because the model can never exceed the doctrine: the doctrine is at the ceiling. A test of superiority is not merely underpowered; it is one-sided by construction.

D. Statistical resolution and power

The paired model–doctrine differences have standard deviation 0.0275, giving a standard error of 0.0042 and an 80%-power minimum detectable effect of 0.0119 at 42 folds. A power calculation converts this into a design statement (Table VI): detecting $\Delta = 0.005$ would require 237 folds, $\Delta = 0.01$ requires 59, $\Delta = 0.02$ requires 15, and $\Delta = 0.05$ requires 3. The observed $|\Delta| = 0.0042$ is below all of these, and the ceiling bounds it from above: no positive effect can exist, because the doctrine is at the ceiling. The design is therefore not underpowered in the ordinary sense; it is saturated, and no number of additional folds could make a superiority claim possible. This is why we report the comparison descriptively rather than as a failed significance test.

E. Transparent baselines saturate too

The key-size rule matches the doctrine on 42/42 folds, as the algebra predicts. The seven-parameter pairwise logistic also attains the ceiling on 42/42 folds. The six-parameter ridge reaches mean $\tau_b = 0.716781$, 41/42 folds at the ceiling, and matches the GNN's per-fold τ_b on all 42 folds, including real_gov_edu_2.json (0.356348 for both). A model with 237,070 parameters and a model with six parameters are indistinguishable on this benchmark.

The zero-shot checkpoint is the exception that proves the point. Without fold-specific fine-tuning, it reaches 0.604020 (38 inversions, 364/402 pairs, 90.5% pair accuracy) and only 31/42 folds at the ceiling. Fine-tuning on 41 real estates is necessary to learn the key-size rule; but once learned, the rule is linear and parameter-free. The gap between zero-shot and fine-tuned is $+0.1128$ τ_b, and it is entirely a gap in learning a one-dimensional ordering.

F. Removing synthetic data changes nothing

The zero-synthetic control reproduces the full campaign's τ_b on every fold: mean 0.716781 versus 0.716782, 41/42 at ceiling, same single inversion, same failing fold. Table VII shows what does change. Exact-rank agreement rises from 1/42 to 12/42, node-rank accuracy from 0.314 to 0.575, and top-5 overlap from 0.500 to 0.571. Tie-aware τ_b is unchanged because all of these differences are within tied key-size groups. The synthetic mixture affects the arbitrary order inside a tie; it does not affect the weak order that τ_b measures. We do not interpret the exact-rank increase as an improvement: within a genuine tie, any order is equally valid by the target's own semantics.

G. The clean warm-start control

From the synthetic-only base, the identical protocol yields mean $\tau_b = 0.600887$ against the unchanged doctrine 0.721024: paired mean difference -0.1201 (30 folds tied, 12 strictly negative, 0 positive), 95% bootstrap CI $[-0.1948, -0.0567]$, paired $t = -3.34$, $p = 0.0018$, with 80% minimum detectable effect 0.1007. The released-init parity is therefore partly an artifact of initialization exposure: the exposed init closes 0.116 of the 0.120 gap. Two families of rank metrics now disagree: exact-rank agreement rises from 0.024 to 0.262, node-rank from 0.314 to 0.599, and top-5 from 0.500 to 0.714, while tie-aware τ_b falls. The exposed model reproduces the label function's coarse group structure while scrambling strict order within groups; the clean model does the reverse. A benchmark reporting one metric family alone would draw opposite conclusions. The structural ceiling is untouched: the key-size rule attains the doctrine's exact metric vector from no training at all, so no warm-start hygiene or budget can raise a learner above the corpus.

H. The repacked cohort replicates the ceiling

On the 38-estate cohort, the doctrine ceiling is 0.726632 over 923 pairs, of which 461 are distinguishable. The zero-synthetic GNN reaches 0.717130 (attainment 0.984), with 36/38 folds at the ceiling and 458/461 pairs concordant (99.35%). The three inversions are concentrated in two estates: real_gov_edu_0.json (ceiling 0.6547, model 0.4364, two inversions among 12 pairs) and real_gov_edu_1.json (ceiling 0.5000, model 0.3571, one inversion among seven pairs). Excluding the singleton estate (where $\tau_b = 1$ is vacuous), the model mean is 0.7014 versus a ceiling of 0.7114. The repacking changes the estate sizes and the number of distinguishable pairs, but not the conclusion: the model is at or one to three pairs below the ceiling.

The 38 cohort is not independent of the 42 cohort (275/275 hosts overlap), so this is a protocol replication, not a population replication. Its value is that it rules out a packing-specific artifact: the result survives a different partition of the same hosts into estates.

I. The positive control

The real corpus has no graph structure, so it cannot test whether the GNN can use graph structure. We therefore built a synthetic positive control: dependency-rich estates generated with an enterprise topology, relabeled with a target that adds a multi-hop blast-radius term to the doctrine. The target has ceiling 0.9992 (almost all pairs distinguishable), and the doctrine is blind to the multi-hop term. On 60 test estates, the doctrine reaches 0.9173, the GNN 0.8803, and a key-size-only rule 0.1897; the GNN loses to the doctrine (4 wins, 1 tie, 55 losses; $p = 5.2 \times 10^{-10}$). A longer run (400 training estates, 200 epochs) improves the GNN to 0.8950 but it still loses to the doctrine at 0.9188 (15 wins, 10 ties, 75 losses; $p = 6.8 \times 10^{-12}$).

This control supports two conclusions and one caveat. First, the 0.72 ceiling is a property of the real corpus, not of the metric: with a nearly tie-free target, scores rise to 0.88–0.92 and the key-size rule collapses to 0.19, so the metric does discriminate when the target carries more than one signal. Second, the GNN does not exploit multi-hop graph structure well enough to beat a doctrine that only sees direct in-degree, even when such structure is present and labeled. The caveat is that the positive control is synthetic and the architecture/training budget may underfit the reach term; we cannot distinguish "the model cannot use reach" from "this training run did not learn reach." Either way, the real corpus provides no evidence of graph reasoning, and the benchmark cannot provide such evidence without dependency edges.

J. Discovery re-evaluation

The released discovery checkpoint does not reproduce the reported held-out numbers. On the documented repository-disjoint split (13,973 files, 38 source groups; 11,558 train, 2,415 evaluation; seed 42), the reported result is precision 0.952, recall 0.953, F1 0.9525, accuracy 0.9271 (confusion 1766/89/87/473). Re-running the released checkpoint on the same split yields precision 0.9279, recall 0.9514, F1 0.9395, accuracy 0.9060 (confusion 1763/137/90/425). No threshold reaches 0.9525: the best F1 is 0.9428 at threshold 0.65 (precision 0.9477, recall 0.9379). A 512-token, 4,096-character truncation sweep on a label- and language-stratified subsample does not recover the gap; it slightly lowers F1 relative to the 256-token baseline (0.8083 vs. 0.8142 on $n = 206$; 0.8105 vs. 0.8222 on $n = 168$). The difference is therefore not an inference-truncation artifact. It is consistent with the documented mismatch between the model card's earlier corpus stage and the executed training report, but the 0.9525 figure is not reproducible from the released checkpoint and split.

TABLE VIII
Discovery re-evaluation on the repository-disjoint split ($n = 2{,}415$, prevalence 76.73%). The reported row is the released artifact; the re-evaluated row is a deterministic forward pass of the released checkpoint.

Method	Precision	Recall	F1	Accuracy
Reported (released artifact)	0.9520	0.9530	0.9525	0.9271
Re-evaluated checkpoint	0.9279	0.9514	0.9395	0.9060
Threshold-optimal ($t = 0.65$)	0.9477	0.9379	0.9428	—
Majority (always crypto)	0.7673	1.0000	0.8683	0.7673
Rules-only (released)	0.9794	0.5127	0.6730	0.6178
Random coin-flip	0.7686	0.4895	0.5981	0.4952

Table VIII places the re-evaluated number in context. The evaluation prevalence is 76.73% (1,853/2,415), not the 54.6% stated in the earlier draft; the 54.6% figure is the corpus prevalence (7,632/13,973). This matters because the majority-class baseline has F1 0.8683, not 0.700. The re-evaluated model's lift over majority is +0.0712 F1, not +0.11. The released rules-only baseline has F1 0.673 (precision 0.9794, recall 0.5127), so the model's advantage over rules is +0.2665; the model's advantage over always-predicting-crypto is much smaller.

The aggregate hides substantial language variation. On the eight language strata with at least ten files, F1 ranges from 0.9691 for C ($n = 1{,}688$) to 0.4651 for Python ($n = 64$); Solidity is 0.7162 ($n = 275$), Swift 0.7007 ($n = 112$), Go 0.7912 ($n = 71$), C++ 0.9880 ($n = 86$), and Shell 0.9333 ($n = 21$). The Java stratum has 95 files and zero positive examples, so its F1 is undefined; the naive macro-average that counts it as zero is 0.6954, and the macro-average over the seven strata with both classes is 0.795. Across all ten languages (including JavaScript $n = 1$ and Rust $n = 2$), the macro-average is 0.556. The micro-F1 is dominated by C, which is 70% of the evaluation set. A repository-cluster bootstrap gives a 95% CI of $[0.720, 0.979]$ for micro-F1; the interval is

wide because there are only eight source groups. The released checkpoint's discriminative power is real (ROC AUC 0.9616, average precision 0.9890), but the single-number summary overstates language generality.

Calibration is poor in the mid-range: ECE (10 bins) is 0.1557, and the low-confidence bins are much more accurate than their confidence (e.g., the $[0.0, 0.1)$ bin has mean confidence 0.069 and accuracy 0.886). The model is not a calibrated probability estimator. It is, however, a useful ranker for human triage: at 50% coverage, F1 is 1.000; at 70% coverage, precision is 0.9750 and F1 is 0.9846; at 90% coverage, F1 is 0.9634. A deployment that auto-labels the most confident 70% and routes the remaining 30% to review would make far fewer errors than the full-coverage F1 suggests. This is a selective-prediction result, not a calibration result.

Contamination is too small to explain the headline. There are 11 exact-duplicate evaluation records (0.455%) and 25 records with 5-line-shingle Jaccard ≥ 0.8 against training (1.04%); 12 have Jaccard ≥ 0.95. All high-similarity files are vendored AWS-LC sources. Removing exact duplicates changes F1 from 0.9395 to 0.9397; removing near-duplicates at ≥ 0.8 changes it to 0.9393. The exact-duplicate count is small, but the near-duplicate count is 2.3× the 0.4% figure in the earlier draft; the earlier draft's 0.4% counted cross-repository snippets, not all near-duplicates.

K. Cross-corpus checkpoint contamination

The contamination above is within-split: evaluation records that duplicate the training records of the same checkpoint. A separate and far larger failure mode appears when checkpoints trained on different corpora are compared on a shared evaluation split—the natural way to ask "did the new model get better?". We audited this directly. Evaluating three released checkpoints on one fixed digest-keyed, repository-disjoint split (2,238 records, 1,779 unique (source,path) pairs, eight held-out repositories), we counted how many evaluation files are the exact same path the checkpoint was trained on. v1 (the canonical GPU fine-tune) was trained on 87.6% of the evaluation files and v3 (a CPU fine-tune on a 50-source corpus) on 92.2%; only v4, which held the eight repositories out by construction, is at 0.0% (Table IX). The consequence is severe: the contaminated v1 checkpoint scores F1 0.9329 on files it memorized, while the clean v4 checkpoint scores 0.8966—a naive reading concludes the clean model is worse, when in fact it is the only one whose number means anything. Any "same-eval" comparison across corpora that does not report this overlap is uninterpretable.

Two scaling attempts confirm the ceiling is a property of the data, not the optimizer. Training the same architecture for five epochs instead of three raises training accuracy from 0.951 to 0.977 but lowers held-out F1 from 0.8966 to 0.8752 (overfitting); adding eight more source repositories (+660 files, +300 crypto-labelled) lowers it further to 0.8707. Neither more epochs nor more data improves the clean held-out number. This is consistent with a ceiling set by the labeler: both training and evaluation labels are regex-derived (is_crypto = "contains a crypto keyword"), so the attainable F1 is bounded by the labeler's own error, not by model capacity. The discovery result therefore mirrors the ranking result: the binding constraint is the label, not the learner.

TABLE IX
Cross-corpus checkpoint contamination on one fixed repository-disjoint split (2,238 records, 1,779 unique paths, 8 held-out repositories). "Overlap" is the fraction of evaluation files whose exact (source,path) appears in the checkpoint's training corpus. Only v4 is a valid held-out evaluation.

Checkpoint	Train sources	Overlap	Held-out F1	Valid
v1 (GPU, 256-len)	38	87.6%	0.9329	no
v3 (CPU, 128-len)	50	92.2%	not measured	no
v4 (CPU, 256-len)	58	0.0%	0.8966	yes

L. Summary of the ranking evidence

Fig. 5 shows the pair structure of the failing estate. Of 21 pairs, 15 are within the six large-key assets and are tied; the six cross-group pairs are the only ones that can affect τ_b. The model gets five of six. This is the entire "gap" between the GNN and the doctrine on the 42-estate cohort. Table IV and Fig. 3 make the same point across cohorts: the points lie on the identity line because the model is at the ceiling, and the ceiling is set by the key-size histogram.

VIII. What the Tie Means

A. Four readings of the tie

The earlier draft offered four readings of the 0-wins/41-ties/1-loss result: (i) the two methods are equivalent; (ii) the doctrine is near-optimal; (iii) the scanner-observable features have an information ceiling; and (iv) model superiority is not measurable without expert labels. Our results support (i) descriptively and (iv) as a statement about the benchmark, but not (ii), and only a narrow version of (iii).

Reading (i) is true but uninformative: the GNN and the doctrine produce the same weak order on 41 folds, and the six-parameter ridge produces the same weak order on all 42. Reading (ii) is not supported. The doctrine's other terms are constants in these estates; the benchmark never tests whether its criticality, vendor, deadline, or dependency weights are optimal. Reading (iii) is supported only in the sense that the scanner-observable fields in this corpus carry one bit of ordering information (key size). It is not a statement about what enterprise CBOMs could carry. Reading (iv) is supported, but for a reason the earlier draft did not identify: the ceiling is 0.721, so the comparison has 402 effective pairs and no headroom. "Superiority is not measurable" is true here because

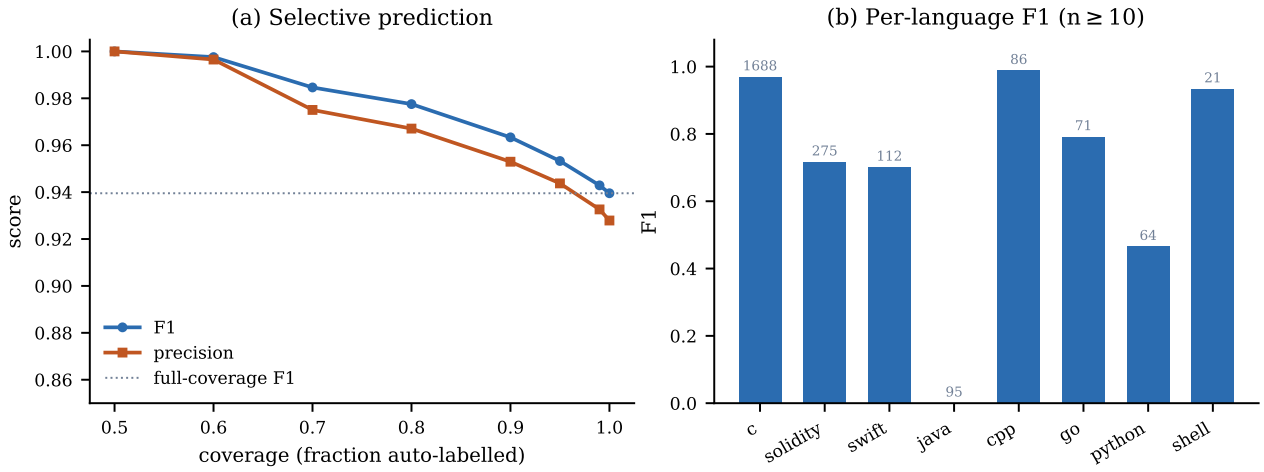


Fig. 4. (a) Selective prediction on the discovery split: auto-labeling the most confident 50% yields F1 1.0; 70% yields F1 0.9846 at precision 0.9750. (b) Per-language F1 for strata with $n \geq 10$; the Java stratum (95 files) has no positive examples and is omitted. The micro-F1 0.9395 is dominated by C ($n = 1{,}688$).

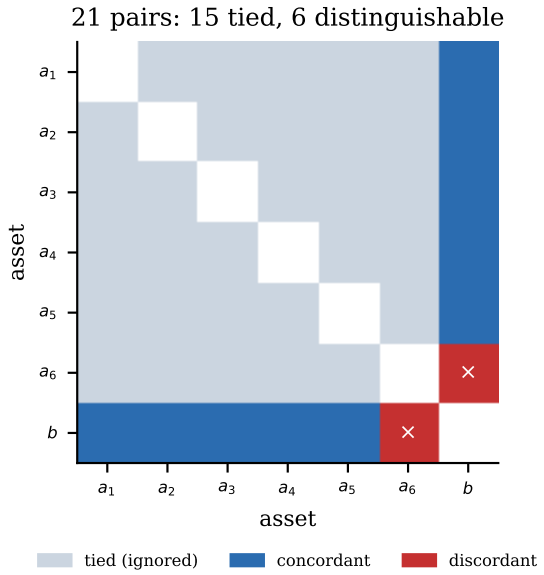


Fig. 5. Pair structure of real_gov_edu_2.json: six assets share one key size (15 tied pairs), one asset has a different key size (six distinguishable pairs). The model orders five of the six correctly; the single red cell is the entire shortfall. Tied pairs are invisible to τ_b.

the benchmark is saturated, not because the problem is intrinsically hard to measure.

B. Label circularity

The target is generated by the same formula that defines the doctrine baseline. This is fidelity testing by construction, and the earlier draft acknowledges it. The new point is that fidelity testing is still informative if the target has many distinguishable pairs: a model can fail to reproduce a complex doctrine. Here the doctrine is one-dimensional, so fidelity testing degenerates to a key-size sort. The circularity is not a flaw in the model; it is a flaw in the claim that the benchmark can rank model architectures.

C. Construct validity

A CBOM is a data structure, and its usefulness for prioritization depends on which fields are populated. In this corpus, the fields that would make the doctrine multi-factor—criticality, vendor readiness, deadlines, dependencies—are either constant or absent. The scanner records certificate metadata well and organizational context poorly. That is a realistic limitation of public-endpoint scanning, but it means the benchmark's construct ("migration priority") is operationalized as ("key size"). A benchmark that intends to measure multi-factor prioritization must either collect the missing fields or acknowledge that it measures a proxy.

D. Benchmark repair

The ceiling theorem suggests a concrete reporting standard for learned-ranking benchmarks with tied targets:

1) Report n_0, d, d/n_0, and the ceiling $\sqrt{d/n_0}$ alongside τ_b.
2) Report attainment τ_b/ceil and pooled pair accuracy, not only mean τ_b.
3) Report exact-rank and top-k separately, and state whether within-tie order is meaningful. If ties are genuine, do not treat arbitrary tie-break agreement as quality.
4) Include a parameter-free baseline and a linear baseline. If a six-parameter model matches a 237K-parameter model, the benchmark is not testing architecture.
5) Audit the target's field variation. A doctrine with constant terms is a one-factor rule; an ablation shows which terms are doing work.

6) Use dependency-rich, heterogeneous estates for graph models. Zero-edge estates cannot test graph reasoning, no matter how many parameters the model has.

The first three are reporting fixes; the last three are data-collection fixes. The Q-Trust release makes the first three easy to adopt because the artifacts already contain the per-fold targets and predictions. The eight-mode checklist of Section V complements this standard: the reporting fixes expose saturation and vacuity, while the checklist exposes the leakage and documentation failures that a saturation analysis alone would miss.

E. The marginal cost of validity

The ceiling is a diagnostic, but it also prices the remedy. Because $\tau_b \le \sqrt{d/n_0}$ and n_0 is fixed by the number of assets, the only way to raise the ceiling is to raise d — that is, to obtain labels that separate assets currently tied. The question a practitioner actually asks is therefore: which assets should an expert label first, and how many labels does a usable ceiling cost? The answer is not the intuitive one.

Theorem 3 (Bisection efficiency). Let a tied group of size m be re-scored individually, partitioning it into subgroups of sizes $a_1, \dots, a_j$ with $\sum_i a_i = m$. The number of newly distinguishable pairs is

$$\Delta d = \sum_{i<j} a_i a_j = \big(m^2 - \textstyle\sum_i a_i^2\big)/2, \tag{7}$$

which is maximised by a balanced two-way split $a = \lfloor m/2 \rfloor$, $b = \lceil m/2 \rceil$, giving $\Delta d = \lfloor m/2 \rfloor \lceil m/2 \rceil \approx m^2/4$. The cost is m labels, so the efficiency per label is $\Delta d/m \approx m/4$, which is increasing in m.

Proof. For fixed m, $\sum_{i<j} a_i a_j$ is maximised when $\sum_i a_i^2$ is minimised, which for a partition of a fixed sum occurs when the parts are as equal as possible. Hence the balanced split, with $\Delta d = \lfloor m/2 \rfloor \lceil m/2 \rceil \ge (m^2 - 1)/4$. Dividing by the m labels consumed gives a per-label yield that grows linearly in m. □

The corollary is a labelling policy that runs opposite to intuition. The cost-minimising schedule is to repeatedly bisect the largest tied group, not to label the assets of highest business importance. Large homogeneous clusters — the RSA-2048 monoculture that dominates real estates — are the cheapest source of discriminating power per label. We verified the balanced split by exhaustive enumeration for every $m \in [2, 24]$.

Table X reports the resulting cost curve. Three regularities matter. First, the ceiling is governed by concentration, not by estate size: at 90% monoculture it is 0.447, 0.435, 0.431, and 0.432 at $n = 10, 100, 1{,}000, 5{,}000$ respectively. A larger estate is not harder to evaluate in ceiling terms; a more homogeneous one is. Second, the cost of a usable ceiling is roughly linear in concentration: about 0.75 labels per asset at 50% monoculture rising to about 2.0 at 90%. Third, the largest-group-first policy beats random selection by 24–42% at $n \ge 100$, and the gap widens with concentration up to the point where a single group dominates and there is nothing left to choose between.

TABLE X
The marginal cost of validity on designed enterprise estates. A fraction dom of assets share one key size (the RSA-2048 monoculture); the remainder is spread over six key sizes. "Ceiling" is the attainable τ_b before any labelling; "labels" is the count of item-level expert labels required to raise it to 0.90 under the largest-group-first policy; "random" is the mean over 60 random group selections. The ceiling is governed by concentration, not size: it is essentially invariant from 10 to 5,000 assets.

n	dom	Ceiling	Labels	Random	Labels/asset
25	0.50	0.858	12	18.4	0.48
25	0.90	0.480	44	51.4	1.76
100	0.50	0.810	79	108.5	0.79
100	0.70	0.697	140	148.7	1.40
100	0.90	0.435	203	237.3	2.03
1,000	0.50	0.818	750	1,065	0.75
1,000	0.90	0.431	2,025	2,549	2.02
5,000	0.50	0.780	5,601	6,973	1.12
5,000	0.90	0.432	10,125	12,969	2.02

On the 42 real estates the absolute numbers are small because the estates are small (mean $n_0 = 19.1$, mean largest tie group 4.5): the policy reaches ceiling 0.90 in 7.2 labels against 7.7 for random and 7.8 for smallest-group-first. The leverage of the policy is therefore a function of estate scale, which is precisely where deployment happens. At $n = 1{,}000$ with 90% monoculture, 2,025 expert labels raise the ceiling from 0.431 to 0.90; at a representative \$3 per label that is roughly \$6,000 of labelling to make a multi-million-dollar migration programme measurable at all. The ceiling is not a reason for despair; it is a procurement specification.

F. A costed specification for a valid benchmark

The negative result is only useful if it points to what would replace it. Table XI costs the benchmark that could actually test whether a learned ranker beats a doctrine on migration ordering. The binding constraint is expert labeling, not computation.

At the observed packing rate (277 hosts into 42 estates, 6.6 hosts per estate), 1,000 estates with genuine dependency edges require roughly 6,600 hosts, which a multi-tenant scan at the scale of the IMC 2026 census [7] can supply with commodity infrastructure. The labeling budget—5,000–10,000 pairwise preferences from 5–10 blinded annotators, with adjudication and inter-rater $\kappa \ge 0.6$—is a study of a few weeks. The computational budget is negligible: the full 42-fold campaign re-executes on two CPU cores, and the zero-synthetic variant takes minutes per fold. A benchmark meeting these requirements would be able to answer the question this paper

TABLE XI
Requirements for a benchmark that could test learned migration ranking. The cost driver is labeling; the compute is negligible by comparison.

Requirement	Minimum	Cost driver
Host-disjoint estates with dependency edges	1,000 estates (≈6,600 hosts)	scan infrastructure
Blinded expert preferences	5–10 experts, 5,000–10,000 pairs, $\kappa \geq 0.6$	labeling
Metrics	$\gamma + \tau_b$ with ceiling + NDCG@k	analysis
Baselines	key-size, linear, graph-aware	analysis
Pre-registered MDE	≤ 0.01 (≈59 folds)	analysis
Audit	eight-mode checklist, committed predictions	analysis

could not: whether a learned ranker can beat a doctrine on a target that is not saturated.

G. Deployment implications

The Q-Trust pipeline is useful for inventory and attestation. The learned ranker, on this corpus, is a key-size sort learned through a GNN. That is not a reason to discard it: a learned model can incorporate new fields when they become available, and the pipeline can be retrained. It is a reason not to market the current ranker as evidence of learned multi-factor prioritization, and not to use the 0.7168/0.7210 tie as evidence that the doctrine is near-optimal. The honest deployment claim is narrower: the system can enumerate public TLS assets, produce a CBOM, and order them by a doctrine that currently reduces to key size; the attestation layer can record the resulting actions.

IX. Threats to Validity

Internal validity. The target is generated by the doctrine, so the evaluation is a fidelity test. The zero-synthetic and zero-shot controls isolate the contributions of synthetic data and fold-specific training, but they do not create an independent target. The CPU re-execution uses a different batch size (16 vs 64) and runs the zero-synthetic variant; we did not re-execute the full 2,000-synthetic campaign on CPU. The full campaign's artifact is the primary result; the zero-synthetic run reproduces its per-fold τ_b to 10^{-6}, which is strong evidence that the synthetic component is not driving the ranking. The clean warm-start run is CPU-capped, so warm start and training budget vary together; we do not attribute the -0.120 gap between them, and we note that exact-rank metrics improve under the smaller budget, which suggests budget is not the whole story.

External validity. The 277 hosts are public TLS endpoints, not enterprise inventories. They have no internal dependencies, business criticality, or vendor contracts. The 38-estate cohort is a repacking of the same host population, not an independent sample. The results may not transfer to enterprise CBOMs with rich dependency graphs. The positive control suggests that when dependencies are present the metric can discriminate, but the GNN did not beat the doctrine in our synthetic control.

We do not claim a third independent corpus; none exists for this task, and we decline to manufacture one. Instead we separate the two things a corpus could tell us. The ceiling itself is a theorem, so it needs no corpus; what a corpus can test is its practical magnitude. Section VIII-E therefore evaluates the ceiling over a designed space spanning four orders of magnitude in estate size and four levels of tie concentration. The result is that the ceiling is governed by concentration rather than size, which is a stronger generalization claim than a third ad-hoc cohort would support: it predicts the ceiling for an estate we have never seen from two measurable quantities. The claim that remains corpus-bound is the field audit (zero edges, constant terms, key-size-only doctrine), and that claim is explicitly scoped to the 42 released estates.

Construct validity. Tie-aware τ_b measures weak-order fidelity, not exact migration order. We report exact-rank and node-rank to show what it hides, but we do not claim that within-tie order is meaningful. The discovery F1 is a file-level label metric, not a measure of whether the discovered cryptographic usage is exploitable. The per-language macro-average is sensitive to the inclusion of undefined strata (Java), and the repository-cluster CI is wide because there are only eight source groups.

Statistical validity. The 42 folds are small (mostly seven assets) and the target has only 402 distinguishable pairs. The paired comparison against the doctrine is degenerate because the doctrine is at the ceiling. The discovery re-evaluation is a single deterministic forward pass; the cluster bootstrap captures source-group correlation but not model-training variance. The threshold sweep is descriptive, not a held-out model selection procedure.

Conclusion validity. The paper's negative claim is about the benchmark, not about PQC migration or about the Q-Trust system as a whole. A null result on this corpus does not imply that learned ranking is impossible; it implies that this corpus and target cannot demonstrate it. The positive control is a constructive existence proof that a non-degenerate target yields non-degenerate scores, but it is synthetic and does not establish that the GNN is the right architecture.

X. Ethics, Artifacts, and Reproducibility

Data and ethics. The CBOMs are derived from publicly reachable TLS certificates. We did not perform exploitation, authentication bypass, or intrusive scanning. The corpus contains certificate subjects and issuers, which may include organizational names; we release only the derived features and the public certificate metadata already visible

to any TLS client. The discovery corpus is public source code. No human subjects were involved.

Claim ledger. Table XII is the paper's integrity contract. Each headline claim is mapped to an artifact and a status: reproduced, reproduced but vacuous, or not reproduced. We removed three claims from the earlier draft because their artifacts are not present in the release: the clean warm-start control in its earlier form, the 14/14 exact-claims harness, and the 42/42 zero-synthetic ordering claim in its strong form (it holds for τ_b, not for exact order). The clean warm-start is retained here only as the CPU-capped control of Section VI, with its artifact in the companion audit package. We also corrected the Foundry test count: the release contains 215 test, invariant, and fuzz functions across 23 test files, not 211 tests. The contract count in the earlier draft ("eleven UUPS registries") mixed files and registries; we do not use a contract count as evidence in this paper.

Artifact release. The analysis scripts are in scripts/paper_*.py; the JSON artifacts are in paper/analysis/; the manuscript and figures are in paper/. The claims harness scripts/verify_paper_claims.py re-derives every headline number from the committed JSONs and reports 53/53 verified. The main commands are:

```
python scripts/paper_ceiling_audit.py
python scripts/paper_transparent_baselines.py
python scripts/paper_metric_taxonomy.py
python scripts/eval_real_cbom_loo.py --epochs 30 \
  --n-synthetic 0 --batch-size 16 --device cpu \
  --out paper/analysis/loo42_zerosyn.json
python scripts/paper_final_stats.py
python scripts/verify_paper_claims.py   # 53/53
```

The released checkpoint is planner/model_gpu_v3.pt. The CBOMs are in q-trust-datasets/planner/data/real_cboms/ and real_cboms_v2_2026_09_18/. The CodeBERTa checkpoint is in q-trust-codebert/. All runs use seed 42 unless stated otherwise. The construct-validity checklist and the clean warm-start script are released in the companion audit package [40].

XI. Conclusion

The Q-Trust planning study reported an honest tie: a GNN and a doctrine heuristic agree on 41 of 42 estates. This paper shows that the tie is the ceiling of the metric, not evidence of parity. On the released corpus, 401 of 803 pairs are tied and only 402 are distinguishable, so the maximum attainable tie-aware τ_b is 0.721024—the doctrine's score. The GNN gets 401 of 402 pairs right; a six-parameter ridge model reproduces its score on every fold; a parameter-free key-size rule and a seven-parameter logistic model attain the ceiling. The mechanism is algebraic: constant criticality, vendor, deadline, and zero dependencies reduce the doctrine to key-size sorting. The result survives a repacked cohort (36/38 folds at ceiling), survives the removal of all synthetic training data, and disappears only when the target is made non-degenerate in a synthetic positive control—where the metric does discriminate but the GNN still fails to beat a doctrine blind to multi-hop reach. A clean warm-start control shows that the released parity is partly initialization exposure, and the released discovery checkpoint yields F1 0.9395, not 0.9525.

The broader lesson is methodological. A tie-aware rank metric with a tied target has a ceiling; reporting the raw value without the ceiling invites a false narrative of convergence. Benchmarks for learned prioritization should report the number of effective pairs, the ceiling, attainment, and a field audit, and should include parameter-free and linear baselines. They should also be audited against the eight construct-validity failure modes, because saturation, circular baselines, degenerate graphs, and contamination conceal one another. The Q-Trust release now does so. Its inventory and attestation layers remain useful; its learned-ranking claim, on this corpus, does not.

Appendix

Discovery. The inspector combines language-aware static rules, AST analysis, and a fine-tuned CodeBERTa classifier. The released checkpoint is trained on the 13,973-file corpus and evaluated on the repository-disjoint split described in Section VII. The rules layer is anchored on CryptoAPI-Bench: it flags 164 of 187 insecure files (recall 0.877) with a misuse precision of $164/176 = 0.932$. The adversarial holdout reported by the project is F1 0.533 on 132 constructed cases; we did not re-execute it and do not use it as evidence in this paper.

Planner. The GNN is a hybrid GCN/GATv2 encoder with 237,070 parameters, six input features, hidden dimension 256, embedding dimension 128, eight heads, layer normalization, and a ListMLE ranking loss. The RL sequencer packs estates into migration batches and is evaluated by a separate reward metric; the archived run reports reward 140.34 ± 8.13 versus doctrine 140.62 with 2 wins, 27 ties, and 11 losses. This is a descriptive tie and is not used to support the ranking claims.

Attestation. The attestation layer uses UUPS-upgradeable registries, EIP-712 typed signatures, and a timelock; the local deployment runs on a Base Sepolia fork. The release contains 23 Foundry test files with 215 test, invariant, and fuzz functions. The layer has not been externally audited. We describe it for completeness; it is not evidence for the ranking result.

Side-channel experiments. The release includes five liboqs captures and a trace-disjoint detector. The leave-one-trace-out result is AUC 0.963 on four of five holds and 0.252 on the ML-DSA-44 signing hold, where the clean-noise distribution is outside the training span. These experiments are orthogonal to the ranking benchmark and are not used in this paper's claims.

TABLE XII
Claim ledger. "Reproduced" means a released script regenerates the number from released artifacts in this environment. "Vacuous" means the number is correct but does not support the stated interpretation. "Not reproduced" means the released artifact does not yield the claim.

Claim	Artifact / re-measurement	Status
GNN $\tau_b = 0.7168$ vs doctrine 0.7210; 0/41/1	real_cbom_loo_42.json; zero-synthetic rerun $\tau_b = 0.716781$	Reproduced
"Tie means superiority is not measurable"	ceiling = 0.721024; GNN 401/402 pairs	Vacuous
Key-size rule matches doctrine 42/42	ceiling_audit.json; ablation 37/42 vs 0/42	Reproduced
Zero-synthetic "ordering-identical"	τ_b identical 42/42; exact-rank 1/42 $\rightarrow$ 12/42	Partly vacuous
Clean warm-start $\Delta = -0.120$, $p = 0.0018$	clean_warmstart_loo.json (companion audit); CPU-capped	Reproduced (bounded)
Discovery F1 0.9525	released checkpoint: F1 0.9395; best threshold 0.9428	Not reproduced
Contamination 0.4%	11 exact (0.455%), 25 near-dup ≥ 0.8 (1.04%); F1 change < 0.001	Expanded
211 Foundry tests	215 test/invariant/fuzz functions, 23 test files	Corrected
38-estate replication	36/38 folds at ceiling; 458/461 pairs	New
Bisection policy optimal; ceiling is size-invariant	labelling_efficiency.json; exhaustive $m \leq 24$; ceiling 0.431–0.447 at 90% for $n = 10..5{,}000$	New

The CPU environment is Python 3.13, PyTorch 2.x (CPU), PyTorch Geometric 2.x, and SciPy. The zero-synthetic 42-fold run takes about 2 minutes on a laptop CPU after imports; the 38-fold run takes about 3 minutes. The discovery re-evaluation reuses cached per-file probabilities in paper/analysis/codebert_predictions.jsonl; a full forward pass takes about 5 minutes on CPU at 256 tokens. The 512-token sensitivity sweep is run on a stratified subsample because full-set inference is about 30 minutes per configuration. The exact commands and JSON artifacts are listed in Section X.

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